Zaytuna Mid-term

MCQs

Carbohydrate Chemistry

-Which of the following is the parent of all aldoses?

a. Ribose
b. Glucose
c. Glyceraldehyde
d. Dihydroxyacetone

Sucrose is

a. Monosaccharide
b. Disaccharide
c. Polysaccharide
d. Pentose

glucose and galactose are epimers at:

a. C2
b. C3
c. C4
d. C5

In lactose the linkage is

a. α-1,4 glucosidic
b. β-1,4-galactosidic
c. α-1,2-glucosidic
d. β-2,1-fructosidic

The storage form of carbohydrates in animals (animal starch) is

a. Starch
b. Inulin
c. Glycogen
d. Cellulose

Which of the following is the parent of all Ketoses?

a. Ribose
b. Glucose
c. Glyceraldehyde
d. Dihydroxyacetone

One of the following derivatives results from oxidation of monosaccharides

a. Glycerol
b. Glucuronic acid
c. 2-deoxyribose.
d. Sorbitol

Lactose is

a. Monosaccharide

b. Disaccharide

c. Polysaccharide

d. Pentose

Enantiomers (mirror image isomers) are:

a. α-Glucose and β-Glucose

b. Glucose and Fructose

c. Glucose and Mannose

d. D-Glucose and L-Glucose

The anticoagulant GAG (heteropolysaccharide) is:

a. Heparin
b. Hyaluronic acid
c. Chondroitin sulfate
d. Keratan sulfate

The GAG which is responsible for corneal transparency is:

a. Heparin
b. Hyaluronic acid
c. Keratan sulfate
d. Chondroitin sulfate

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All the following are reducing disaccharides, *Except:*

a. Isomaltose
b. Lactose
c. Sucrose
d. Maltose

- Glucose and Fructose are :

a. Functional group isomers
b. Epimers
c. Anomers
d. Enantiomers

- The sugar found in DNA is:

a. Deoxyribose
b. Ribulose
c. Erythrose
d. Ribose

- The sugar found in RNA is:

a. Deoxyribose
b. Ribulose
c. Mannose
d. Ribose

- **The storage form of carbohydrate in plants is:**

a. Starch
b. Cellulose
c. Glycogen
d. Dextrin

- **The only sulfate free GAG is:**

a. Heparin
b. Hyaluronic acid
c. Inulin
d. Dextran

- **The anomers are**

a. D-glucose and L-glucose
b. α-glucose and β-glucose
c. Glucose and mannose
d. Glucose and fructose

- **The lubricant GAG presents in synovial fluid is**

a. Hyaluronic acid
b. Dermatan sulfate
c. Heparin
d. Heparan sulfate

8- The main sugar in animal blood is = grape sugar is:

a. Xylose.
b. Ribose.
c. Glucose.
d. Ribulose.

Lipid Chemistry

Arachidonic acid has 20 carbon atoms with

a. 3 double bonds
b. 2 double bonds
c. 4 double bonds
d. 8 double bonds

Identify the simple lipid from the following?

a. Lecithin
b. Fatty acid
c. Triacylglycerol
d. Steroids

Bile acids are derived from:

a. Cholesterol
b. Amino acids
c. Fatty acids
d. Sugar acids

Which of the following lipids are mostly present in mitochondrial membranes?

a. Waxes
b. Triacylglycerol
c. Cardiolipin
d. Plasmalogen

Which of the following best describes the cholesterol molecule?

a. Simple lipid
b. Compound lipid
c. Derived lipid
d. Neutral fat

Which vitamin is derived from cholesterol?

a. A
b. D
c. E
d. K

The oils are specifically rich in

a. Unsaturated fatty acids
b. Saturated fatty acids
c. Saturated and unsaturated fatty acids
d. None of these

· **Essential fatty acids are**

a. Linoleic acid
b. Arachidonic acid
c. Palmitic acid
d. Oleic acid

The following is a polyunsaturated FA (PUFA)

a. Palmitic
b. Palmitoleic
c. Linoleic
d. Oleic

one of the following is not glycerophospholipid

a. lecithin
b. phosphatidyl inositol
c. Cephalin
d. Sphingomylin

-Sphingosine is *Not* present in

a. Cephalin
b. Sphingomyelin
c. Cerebrosides
d. Sulfolipids

-All the following statements concerning essential fatty acids are correct *Except*

a. They are not formed in the animal body
b. It is essential to take them in diet.
c. Their deficiency produces dermatitis and fatty liver
d. The most essential one is palmitic acid

- prostaglandins are derived from polyunsaturated fatty acids with

a. 16 Carbons
b. 18 carbons
c. 20 carbons
d. 24 carbons

-Prostaglandins are synthesized from

a. Palmitic acid
b. Linoleic acid
c. Oleic acid
d. Arachidonic acid

On complete hydrolysis ,triglycerides yield

a. Glycerol and phosphoric acid
b. Glycerol and a fatty acid
c. Glycerol and two fatty acids
d. Glycerol and three fatty acids

All the following are derivatives from cholesterol except

a. Bile acids
b. Steroid hormones
c. Vitamin D2
d. Bile salts

Carotenoids are precursor for synthesis of vitamin

a. Vitamin A
b. Vitamin D
c. Vitamin K
d. Vitamin E

Snake venom toxin cause death because it contains:

a. Phospholipase A1 enzyme
b. Lecithinase enzyme
c. Lipase enzyme
d. Kinase enzyme

Phospholipid deficiency causes respiratory distress syndrome:

a. Platelet activating factor
b. Cephaline
c. Lung surfactant (dipalmitoly lecithin)
d. Cardiolipin

Protein Chemistry

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1- Which of the following essential amino acids is not synthesized by the body?

a. Arginine

b. Glutamine

c. lysine

d. Proline

4- Which of the following is an essential amino acid containing sulfur?

a. Cysteine

b. Asparagine

c. Valine

d. Methionine

5- Which of the following is an example of imino acid?

a. Alanine

b. Glycine

c. Proline

d. Serine

6- Which of the following amino acids is semi-essential amino acid?

a. Leucine

b. Lysine

c. Isoleucine

d. Arginine

7- Which of the following are simple proteins?

a. Albumins

b. Globulins

c. Collagen

d. All of the above

9- Which of the following are basic amino acids?

a. Lysine and arginine

b. Lysine and asparagine

c. Glutamine and arginine

d. Lysine and glutamine

15- the amino acid in alkaline medium is:

a. positively charged.

b. negatively charged.

c. dipolar ion

d. can't migrate in electric field

18- Age defect in chaperon causes:

a. Fatty liver

b. Alzheimer and Parkinson disease

c. Respiratory distress syndrome

d. Marfan syndrome

20- An aromatic and essential amino acid is:

a. Tyrosine

b. Phenylalanine

c. Proline

d. Arginine

23- Collagen and elastin are:

a. Derived proteins

b. Scleroproteins

c. Phosphoproteins

d. Glycoprotein

24- Denaturation of proteins leads to all of the following, except:

a. Exposure of some hidden groups

b. Increased viscosity

c. Increased solubility

d. Loss of biological activity

26- An example of phosphoprotein is:

a. Caseinogen

b. Collagen

c. Albumin

d. Globulin

28- Which out of the followings is not a fibrous protein?

a. Hemoglobin

b. Collagen

c. Elastin

d. Keratin

29- Which out of the following is not a hemo protein ?

a. Hemoglobin

b. Myoglobin

c. Cytochromes

d. Collagen

32- the bond which stabilize primary structure of protein is:

a. ionic bond

b. hydrogen bond

c. hydrophobic interactions

d. peptide bond

35- α-helix and β-pleated sheets are examples of:

a. Primary structure of protein.

b. Secondary structure of protein.

c. Tertiary structure of protein.

d. Quaternary structure of protein.

Heme Chemistry

2- Heme porphyrin ring contain

a. Copper

b. Iron

c. Zinc

d. Magnesium

3- The hemoglobin which is formed of 2α and 2β chain is:

a. HbA1

b. HbA2

c. HbF

d. HbS

4- Hemoprotein which is present in muscle:

a. Myoglobin

b. Ferritin

c. Transferrin

d. Hemosiderin

7- Carboxy hemoglobin is formed by:

a. Hemoglobin is attached to CO_2

b. Hemoglobin is attached to CO

c. Hemoglobin is attached to oxygen

d. Hemoglobin is oxidized

8- Carbmino hemoglobin is formed by

a. Hemoglobin is attached to CO_2

b. Hemoglobin is attached to CO

c. Hemoglobin is attached to oxygen

d. Hemoglobin is oxidized

9- In HbS glutamic acid is replaced by:

a. Valine

b. Leucine

c. Lysine

d. Alanine

18- HbA_{1c} is used for follow up of patients with:

a. Renal failure.

b. Hepatic failure.

c. Diabetes mellitus.

d. Brain failure.

19- All of the following are normal hemoglobin EXCEPT:

a. HbA_1

b. HbA_{1c}

c. HbS

d. HbF

5- The hemoglobin which is formed of 2α and 2γ is:

a. HbA1

b. HbS

c. HbF

d. Hb C

Enzymes

1- The substrate bind to site of enzyme called:

a) Active site
b) Allosteric site
c) All of the above
d) Non of the above

2- The ascaris worm prevent its digestion by:

a) Antienzymes
b) Allosteric inhibitors
c) Competitive inhibitors

4- The optimum Ph of pepsin is

a. 2
b. 7
c. 8
d. 9

5- As regard Km:

a. It is the substrate concentration which produces Vmax
b. It is the half Vmax
c. It is the half of the substrate concentration which produces Vmax
d. It is the substrate concentration which produces half Vmax

7-Competitive inhibition can be relieved by raising the

a. Enzyme concentration
b. Substrate concentration
c. Inhibitor concentration
d. None of these

18-These enzymes have different structure but the same catalytic function. Frequently they are oligomers made from different polypeptide chains. These enzymes are called:

a. Allosteric enzymes
b. Isozymes
c. Proenzymes
d. Zymogens

9-Sulfonamide:

a. Is required for synthesis of folic acid in bacteria.
b. Acts as competitive inhibitor for synthesis of folic acid.
c. Example of allosteric inhibition to enzymes.
d. Bactericidal drug.

10- Dicumarol and warfarin are:

a. Anticoagulants structurally similar to vitamin K .
b. Examples of non-competitive enzyme inhibitors.
c. Structurally similar to vitamin C.
d. Drugs for treatment of gout.

11- Allopurinol:

a. Is used in treatment of hypouricemia.
b. Inhibits xanthine dehydrogenase.
c. Is used for treatment of Gout.
d. Example of allosteric inhibition.

20-The enzyme which increases in case of myocardial infarction is:

a. LDH 1
b. LDH 2
c. LDH 3
d. LDH 5

21-The enzyme which increases in case of brain infarction is:

a. CK 1
b. CK 2
c. CK 3
d. CK 5

Immune system

1- The immunoglobulin which work in primary immune response is:

a. IgM

b. IgG

c. IgA

d. IgE

3- The only immunoglobulin which can cross the placenta is:

a. IgM

b. IgG

c. IgA

d. IgE

4- The immunoglobulin which work in the secondary immune response is:

a. IgM

b. IgG

c. IgA

d. IgE

5- The main immunoglobulin which is present in secretion is:

a. IgM

b. IgG

c. IgA

d. IgE

9- The most abundant immunoglobulin in blood is:

a. IgM

b. IgG

c. IgA

d. IgE

8- Regarding immunoglobulin one of the following is incorrect:

a) The variable region is present in the pronge of Y

b) The variable region binds to antigen

c) The constant region is present in stem of Y

d) The constant region binds to antigen

Wish you the best of luck